

Lehy-ii (No Energy feedback) Passenger elevators

System Software uses C language

Electrical Debugging Instructions

(Internal Reference)

Version:

English

Total 24 Pages

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1. Debugging Tools

1.1 Screwdriver

Adjust the rotation switch on the plate must use a dedicated screwdriver.

1.2 Repair Computer

For parameter modification and fault diagnosis.

1.3 Flash Rewrite Tool

1) Boot Rom

2) Dual-machine Interconnect Cable (connecting P1 Boards and Computers)

3) Computer ^{*1}

[Note] *1: Installation and use of Boot Rom The corresponding software.

2. Electrical Preparation and confirmation

2.1 Get on the electricity.

- 1) The control cabinet **Auto/hand** (Automatic/Manual) switch dial to **HAND** (manual) side.
- 2) Confirm P1 The wiring of the related connectors on the board is connected correctly.
- 3) Confirm P1 On the board Drsw/rst Switch is Drsw Side to cut off all the door operation requirements.
- 4) Confirm P1 On the board Fwr/mnt Switch isn't in Fwr Side Tca/mca The switch is in the middle bit.
- 5) Confirm rotation Switch "MON1" 、 "MON0"Is" the value of the 1、0.
- 6) Confirm rotation Switch "SET1" 、 "SET0"Is" the value of the 0、8.

2.2 Power on Confirmation

When your control is on the cabinet, P1 Light emitting diodes on the board Ccwdt、Drwdt、Dpok should be lighted; If the door is completely closed, the light-emitting diode41 should be lighted; If the car was in the door area, the light-emitting diode Dz should be lit.

Note The watch timer are used to monitor the plates Cpu is working properly. If you are are in the process of debugging WDT Extinguished, indicating the corresponding Cpu A serious failure has occurred (the program has run away). At this time, need to check the printing plate5V Power wiring and connectors. If Ccwdt Or Drwdt The light flashes, the corresponding program may is successfully written to the flash memory.

3. Manual operation (Low-speed operation)

Manual operation must was done after the power is confirmed.

3.1 Writing of manual pole position

- 1) Close to the power circuit breaker.
- 2) Would P1 The rotary switch on the boardSET1"Tune to"0" , "SET0"Tune to "D" 。
- 3) Would P1 On the board SW1 /sw1 Switch Press to SW1 Side, until 7 Segment Code Display "Pxx" (XX For the number of floors), and began to flicker.
- 4) Run it manually until 7 The segment code shows stop flashing and the pole position is written successfully.
- 5) Press move P1 On the board Rst Toggle the switch and deposit the pole position Rom In
- 6) Would P1 The rotary switch on the boardSET1" 、 "SET0"Revert to 0、8。

<Note>: Enter the corresponding SET The method of the ' is ' to rotate the switch 'SET1" 、 "SET0' To the ' appropriate value, to the ' long press down SW1 Toggle switch, until enter the corresponding mode, then loosen; SET The method of the ' is ' to rotate the switch 'SET1" 、 "SET0" respectively, transferred to 0 and 8, Down Long Press SW1 Toggle the switch until exiting SET mode, then loosen

3.2 Manual Test Run

When the uplink button is pressed, P1 Light emitting diodes on the board 89 and Up Should is lit, the car Should be uplink. Down Button Press, the car should be being down, if the car is heavier than the light, the car would have an upward impact, and t Hen normal downward. The manual run speed should be 20m/min.

During the manual run commissioning, it must be confirmed to the lock can be fully opened and the traction machine no Abnormal noise and vibration. In addition, the "Car stops", the lock contact should be completely closed to ensure the lock.

During the manual operation, the car must be in the safety switch, door switch, car door lock switch and Ul/dl Stop immediately when disconnected.

3.3 Adjustment of weighing device

I write the car no-load weighing value to the memory

- 1) Make sure the ' car ' is unloaded.
- 2) Manually drive the "car" to the "halfway" point of the "the" Road (Refer to the next Page box to confirm the midpoint of the "the" path) 。
- 3) Would P1 The rotary switch on the boardSET1"Tune to"6" , "SET0"Tune to"2" 。
- 4) The sw1./sw1→ switch on the P1 board is pressed to the sw1→ side, the 7-segment code will display the current scale value, and flicker, after writing automatically exits the SET mode, 7 segment codes into the MON display mode.
- 5) The rotary switch on the P1 board "MON1" to "0", "MON0" to "D", Up to the "SW2" switch, 7 section code shows the weight of the no-load car data, to confirm the success of no-load scale write.

The method of confirming,
the car, stops at the
midpoint of the "the" Road

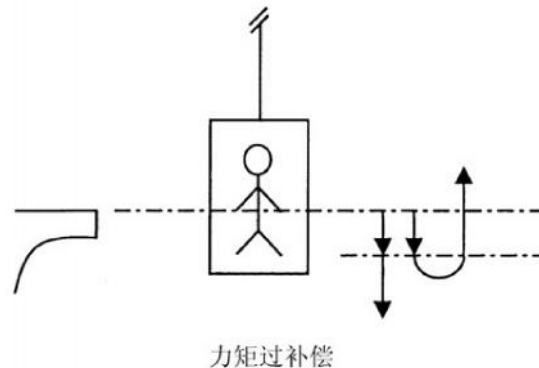
Would P1 The rotary switch on the boardMON1"Tune
to"0" , "MON0"Tune to"1And put the elevator into a
manual operation. Activate the elevator when the
display changes to "0To-0"indicates" the current
position of the ' car ' in the middle position.
Indicates that's distance represented by the256mm。

II Write the car balance load weighing value to the memory

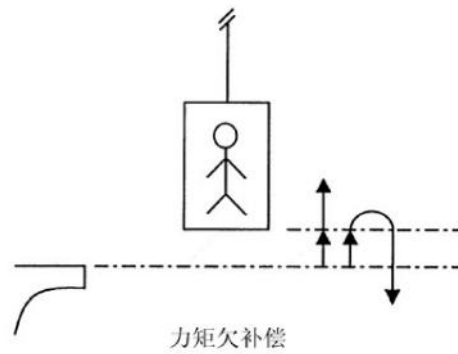
- 1) Confirm is half loaded.
- 2) Manually drive the car to the midpoint of the "the" road.
- 3) Would P1 The rotary switch on the boardSET1"Tune to"6" , "SET0"Tune to"3" 。
- 4) The sw1-/sw1↵ switch on the P1 board is pressed to the sw1↵ side, the 7-segment code will display the current scale value, and flicker, after writing automatically exits the SET mode, 7 segment code into the MON display mode.
- 5) Would P1 The rotary switch on the boardSET1" 、 "SET0"Revert to 0、 8。
- 6) The rotary switch "MON1" on the P1 board is transferred to "0", "MON0" is transferred to "D", the "SW2" switch is pressed down, and the data of the balance load car weight is shown by the 7-segment code, confirming that the balance load scale has been written.
- 7) Would P1 The rotary switch on the boardMON1" 、 "MON0"Revert to 1、 0。
- 8) Press move P1 On the board Rst Toggle Switch to save no-load and balance load scale value Rom In

"Note" Over writing no-load and balance load weighing value, in the Set nameplate marked for backup (NLWT, BLWT)

Iii Fine Adjustment of torque compensation (usually not adjusted)



- 1) On the no-load uplink NL up or no-load downward NL DN In operation, the car has a trend of upward movement when the lock is opened, and the rotating switch "WghgToF"Adjustment, Adjustment method reference 7.2 Section table 2 (SET1 = 3) 。
- 2) On the no-load uplink NL up or no-load downward NL DN In operation, the car has a downward movement trend while the lock is opened, and the rotary switch "WghgTo0"Adjustment, Adjustment method reference 7.2 Section table 2 (SET1 = 3) 。



The above torque compensation adjustment must not as full load **FL** The case.

The following methods can invalidate the torque compensation function.

The rotate switch "Wghg" Tune to "0" .

3.4 Write floor Height in manual mode

- 1) Manual operation would stop the car in the lower terminal layer to the door area.
- 2) Would P1 The rotary switch on the board SET1 "Tune to "0" , "SET0" Tune to "7" .
- 3) Press down P1 On board the SW1 "Switch until the 7 segment code shows" FXX "(xx is the number of floors where the car), and flashing.
- 4) The car from the lowest level of continuous operation to the upper gate area or the upper limit switch (UL). 5 when the "car" at "Top Stop", 7 yards in the "floor to show" elevator floor number and stop flashing.
- 6 If The car stops before reaching, repeat steps 1 ~ 5).

Note

If the layer height parameter cannot be written, check the action position, the flat layer and the device of the Terminal Limit switch.

4. High speed operation

High-speed operation must to performed after manual operation-confirmed

4.1 The rotation switch setting check on the printing board

Name	Set Value	Description
MON1	1	Show elevator Current fault code
MON0	0	
SET1	0	Avoid entering by mistake SET Mode
SET0	8	
Ele.no	-	Parallel and Group Control settings

4.2 Layer Station setting

On/Next Call button Combine Bridging "SET" To carry out the layer station setting.
Set Values to16The in-process format displays, such as:1Floor corresponding Display "1" ,

21stFloor corresponding Display "15" 。 Set Value Display Range:1h~Efh。 The layer station printing plate on the electricity, across the ' plates on the 'SET", enter the floor setting mode. The Dot-matrix display quickly flashes to show the last saved floor setting. When the/When the next Call button is pressed, the floor setting value increases accordingly/Decrease, the increase and decrease of the increment with the duration of the button to maintain the larger; Dot-matrix displays no longer flashing, while the current floor settings are synchronized, and the loop changes when the S Et value is out of range. Once the "button is released," The current setting be displayed with flashes.

Note ' When the ' printed plate is on the ' power ' state, the removal 'SET" The current set value isn't saved until the stitch is across."

4.3 Write floor height in automatic mode

If the layer is being written in the case of automatic operation, the selector will be wrong and the car would be high speed Back to the terminal layer.

- 1) Dock the car to the lower terminal layer.
- 2) Confirm that's elevator car door are in full shutdown condition. (P1 On board 41 Light emitting diode lit)。
- 3) Would P1 On board Drsw/rst Toggle Switch dial to Drsw Side, cut off the door-machine.
- 4) Would P1 The rotary switch on the board SET1 "Tune to 0" , "SET0 "Tune to 7" 。
- 5) Press down P1 On board the SW1 "Switch until the 7-segment code shows" FXX "(xx is the number of floors where's), and flashing.
- 6) The control cabinet Auto/hand(Automatic/Manual) switch dial to AUTO(automatic) side, P1 board 60 The light-emitting diode is lit.
- 7) If the external call on the P1 board are not answered, the elevator would be in a independent mode, and the DCB/FMS switch would be transferred to the DCB side.
- 8) Lift to write layer speed (45m/min) automatically run up, start writing layer.
- 9) When the elevator runs to the upper terminal layer, stop, such as write layer success, 7-segment code display would stop Flashing.
- 10) The elevator runs downward at the writing level, and the purpose of this operation was to verify the level of data just Written.
- 11) When the elevator runs down to the bottom end layer, stop, write layer end, 7 segment code display automatically switch to The MON switch display mode.
- 12) Would P1 The rotary switch on the board SET1" 、 "SET0 "Revert to 0、8。

Note > Unbalanced load data are also written in this procedure

Note

If the layer height parameter cannot be written, check the action position, the flat layer and the device of the Terminal Limit switch.

4.4 High Speed Test Run

- 1) Confirm that's the elevator car door and the floor door are with full shutdown condition. (P1 On board 41 Light emitting diode lit) 。
- 2) Confirm P1 On board Drsw/rst Toggle Switch in Drsw Side, cut off the door-machine.
- 3) Manually park the elevator in the door area near the midpoint of the the the "the" path.
- 4) The control cabinet Auto/hand(Automatic/Manual) switch dial to AUTO(automatic) side, P1 board 60 The light-emitting diode is lit.
- 5) Would P1 On board Tca/mca Switch dial to Mca side, press up or down P1 On board the SW2 "Switch, register uplink or downlink instructions, let the elevator high-speed automatically up or down a layer."

- 6) Confirm 7 Segment code display always shows "000", which is, there are no failure to produce, and the traction machine does not have abnormal noise and vibration."

5. High-speed running debugging

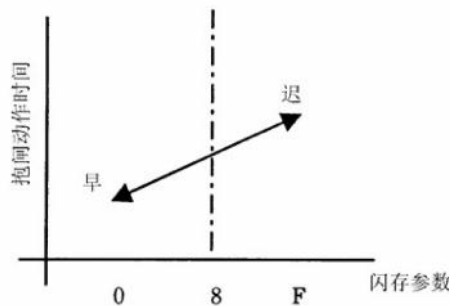
High-speed running debugging must be performed in high speed when the operation is OK

High-speed running and commissioning needs to be adjusted according to ride comfort, leveling precision, Tsd Run checks and load fluctuation checks in sequence.

5.1 Adjustment of Ride Comfort (Generally don't make adjustments)

- 1) LTB (Stop time adjustment of delay lock)

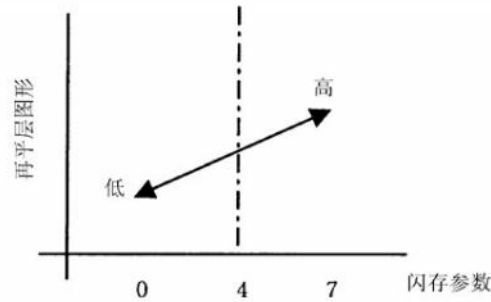
If you are having to adjust, you can use tools such as maintenance computers. As the car stops then the brake release caused by premature stop vibration, you can through the adjustment "LTB" Parameter is resolved. If The delay lock is too short, the parameter is "F" The direction of the adjustment; If too long, to "0" Direction adjustment.



LTB The adjustment

- 2) Rlpt (flat layer graphic constant adjustment)

If the car needs a higher precision to the flat layer, adjust this parameter. If you are having to adjust, you can use tools such as maintenance computer and accelerometer. When adjusting this parameter, the car must be being a balanced load, and the car is moved from the flat layer by manual mode.50mm。 When the elevator was placed in the automatic mode, the car should be being a smooth return to the original Flat-layer. If The leveling process has hysteresis, increase the "Rlpt The parameter 1~2 Units. The adjustment must be very careful, if the increase is much more, the car would too occur.



Rlpt The adjustment

5.2 Adjustment of flat layer precision

1) SHIFT (Residual distance offset parameter adjustment)

If the elevator car in all floors, the downward stop position are "beyond" (up car high, down car lower), then the Parameter will be larger (About 2mm/digit), adjust the method reference 7.2 Section table 2 (SET1 = 1Conversely, if the elevator car

On all floors, the downward stop position is 'insufficient' (the upper car on the lower side, down the ' car are high), then the parameter is reduced.

2) Dnsh (Dzu Position correction parameter Adjustment)

If the elevator car in all floors of the downward stop position are "beyond" (car low), then the parameter will be larger(About 1mm/digit), adjust the method reference 7.2 Section table 2 (SET1 = 2Conversely, if the elevator car in all floors of the downward stop position are "insufficient" (high car), the parameter Will be smaller.

5.3 Tsd Run Check

- 1) Place the car in no-load state and stop to the door area of the non-terminal floor.
- 2) Would P1 The rotary switch on the board SET1 "Tune to"0" , "SET0" Tune to "F", press Down SW1 Until 7 Number of floors displayed by Segment display
- 3) Register on the terminal layer or lower terminal layer call, the car would run along Tsd the graphic stops at the terminal layer.
- 4) TSD action, the elevator will enter the state cannot be restarted, need to be reset operation. The toggle switch "Drsw/rst" on the P1 plate is dialed to the "RST" side for reduction.
- 5) Would P1 The rotary switch on the board MON1 "Revert to"1" , "MON0" Revert to "0"

5.4 Load fluctuation Check

Car Load from 0% Change to 110% and confirm the following requirements.

- 1) Flat Layer Error in ± 10 mm Within.
- 2) There is no noticeable change in ride comfort (especially in the vicinity of the "gate area").
- 3) The amplitude and noise of vibration are within the standard prescribed range.

6. Low Speed Auto Run

If the car because of the safety loop disconnect and emergency stop outside the door, in the safety loop back to normal After the car is the automatic low-speed way to the nearest door to the floor, open the door to achieve the purpose of Rescuing passengers.

Follow the steps below to verify the Low-speed autorun.

- 1) Register the uplink or downlink instructions to enable the elevator to run in high speed.
- 2) When the elevator runs outside of the door area, press the Emergency stop button on the control cabinet, the elevator stops Q Uickly,89 The light-emitting diode is off.
- 3) Make the emergency stop button reset, about a few seconds later#89 The relay would resume suction and enter the low speed automatic to running state.
- 4) The elevator would run in the opposite direction before the original safe loop action.
- 5) The car is in the nearest floor of the door area to stop, and the open the door after the closed standby.

7. P1 On the printed board **Led** (light-emitting diode) function and set switch function

7.1 The Led The function of (light-emitting diode)

Printing Plate	Type	Name	Function	General status
p203758b000	Light-emitting Diodes	Fper	Fpga Error Indication	Fpga Error is lit
		MNT	MNT Switch Action Instructions	MNT The switch is in the middle and the point is lit
		Dn	downlink Instructions	Down Point Light
		Up	Uplink instructions	Up to the point of light
		60	Automatically run instructions	Auto-run Point-in-time Light
		29	Safety Loop Instructions	The safety loop is lit normally.
		89	#89 Relay Action	#89 It ' s a timely light.
		Drwdt	Dr-cpu Work Instructions	Work normally, light up.
		Ccwdt	Cc-cpu Work Instructions	Work normally, light up.
		Dz	Dz Signal Indication	Light up after entering the door area.
		41	41DG Signal Indication	The hall door and the car door are lit after closing.
		Gcok	Group Control Communication Instructions	group-controlled communication is lit at the normal point
		Dpok	Single-ladder Communication Instructions	Single-ladder Communication normal point of light
	7 Segment Code	—	Various directives	Current Fault code ("MON1" = "1" "MON0" = "0")

7.2 The function of each switch

1The function of each switch

Printing Plate		Switch Name	Function	General status
p203758b000 (P1)	Toggle	[Fwr]/[mnt]	Fwr(ON): Allow Flash Write MNT(next): Allow remote transmitter	Put on the middle position when installing
		[Drsw]/rst	Drsw(ON): Cutting Door Machine Rst(next): Elevator reset	Middle position
		Dcb/fms	DCB(ON): Close the door Fms(write layer): Write layer high data	Middle position
		SW1 /sw1	Combination switch	Middle position
		SW2 /sw2	Combination switch	Middle position
		[TCA]/[MCA]	Reference 1	Middle position
	Rotary	SET1	Set Feature Selection ("rotary switch" in detail) SET1" 、 "SET0"Function")	0
		SET0	Set Feature Selection ("rotary switch" in detail) SET1" 、 "SET0"Function")	8
		MON1	Display Feature Selection ("rotary switch" in detail) MON1" "MON0"Function")	1
		MON0	Display Feature Selection ("rotary switch" in detail) MON1" "MON0"Function")	0
		Ele. no	Ladder number Set	

Toggle Switch Name has []The tag indicates that is location can be locked, and no markup indicates is location that cannot.

Note

In addition to rewriting the program, does not dial fwr/mnt to the FWR side at any time.

1 Toggle Switch Tca/mca

Toggle Switch Tca/mca In TCA (上)Position, by up/Press down SW2 Toggle switch, register top floor/The bottom floor instruction makes the elevator run to the terminal; Toggle Switch Tca/mca In MCA (下)Position, by up/Hold the time SW2 Toggle The switch, the elevator will keep up/Run down until it stops at the secondary terminal layer, such as halfway through SW2 Toggle switch, the elevator would dock in the nearest layer.

2) Rotary Switch “MON1” 、 “MON0”function

Table 1 MON Switch function

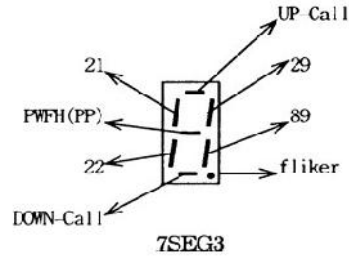
MON1	MON0	Function		7 Segment Code Display Content
0	0	Show elevator status and large class fault reference □2		Elevator state and large classification fault
	1	Show Car Center Location		Car Center Location
	2	Show Sync Layer		Sync Layer
	7	Show forward Layer		Forward Layer
	D	Show scale values		SW2 : no-load Scale Value SW2 : Balance Load Scale value SW2 Middle Level: Current scale value
1	0	Show current minor fault code	Reference Appendix 1	Current small fault Code
	1	Show History Minor Fault Code		Historical small fault Code
3	5	Show Encoder Feedback Speed		Encoder Feedback Speed
	6	Show Percent Scale value		Percent Scale Value
6	0	Set the Clock-年	Reference 3	Year
	1	Set the Clock-月		Month
	2	Set the Clock-日		Few days
	3	Set the Clock-Week		Day of the Week
	4	Set the Clock-时		Hours
	5	Set the Clock-分		Minutes
9	C	Set Manual Maximum Speed	Reference 3	Manual Maximum Speed Code
A	C	Adjust no-load Scale Value	Reference 3	No-load scale Value
	D	Adjust Balance Load Scale Value		Balance Load Scale Value

F	0	Show System Module Version number	Reference 4	System Module Version Number
	1	Show Management Module version number		Management Module Version Number
	3	Show Control Module Version number		Control Module Version Number
	4	Display the driver module version number		Driver Module Version Number
	5	Display the Maintenance module version number		Maintenance Module Version Number
	7	Display parameter Module version number		Parameter Module Version number

2 Elevator state and large classification fault

LED3	LED2	LED1	Large classification Failure
Status Display	E	0	No Fault
		1	Invert side
		2	Rectifier side
		3	Rectifier side
		4	Power
		5	Control Panel
		6	Control Panel
		7	External Safety Loop
		8	Door
		9	Lifting switch
		A	Lock Traction Machine
		B	Speed Control Layer Selector
		C	Scale
		D	P1 Base Plate
		E	System Setup
		F	Communication failure

LED3 the status is shown below:



3 Use MON Methods for setting parameters of switches

Press up SW2, 7 The value that is the segment code displays begins to flicker, and grows, increases to the maximum value, returns the Smallest T value, continues to increase, and so the circulation; Loosen SW2So, it's restored to the median, the display value remains unchanged; SW2, save the current display value to Ram, after a successful save, 7 Segment code shows stop blinking.

For example: 2008 Years 4 Month 18 Day 14 Point 33 Minutes, Friday

Would MON1 Adjusted to 6, MON0 Adjust to 0, 7 The segment code would show "0XX", of the which the Xx" is the last two digits of the number of years, as 2008 Year is displayed "08", the default value is from 2000 Year began("00"), press up SW2Until 7 Segment Code Display "008", let go SW2, and then press down SW2 Save the display value to Ram; According to the same method, the month is set to 04, the date is set to 18, the hour is set to 14, the minutes are set to 33, the week is set to 5

4 Software Version Display Data description

The details of each program are stored in a specific area of the software and can are 7 Segment code displays this information for Example: the software contract # is shown as Ascii Code 8 bytes) ...EAG45901 →

45h, 41h, 47h, 34h, 35h, 39h, 30h, 31h

Date displayed as (BCD, 4 bytes) ... 2006-08-28→

20h, 06h, 08h, 28h

The program name appears (ASCII, 8 bytes) ... sccg01s-a→

53h, 43h, 43h, 47h, 30h, 31h, 53h, 2dh, 41h

[Show examples]

I There is a software contract number

045→041→047→034→035→039→030→031

..... Software Contract Number

Display

→d20→d06→d08→d28

..... Date Display

→r53→r43→r43→r47→r30→r31→r53→r2d→r41 ... Program Name Display
→ (from the software contract number start cycle)

II No Software Contract number

R53→r43→r43→r47→r30→r31→r53→r2d→r41 Program Name Display
→ (start loop from program name)

The displayed content flashes, spaced about 2 seconds. The display is switched the content. If no information is displayed, keep Off state.

3) Rotary Switch "SET1" 、 "SET0"function

Table 2 SET Switch function

SET1	SET0	Function	Description
0	2	Automatic Learning No-load Scale NL	Before entering this mode, the elevator must stop at the lower terminal, the layer would run from the elevator lower A1 layer to the upper terminal layer
	3	Automatic Learning Balance Load Scale BL	Before entering this mode, the elevator must balance the "load to stop" at the bottom terminal, the layer would run From the bottom terminal layer to the upper terminal layer.
	5	Maintenance Scale Write	Use when repairing and maintaining
	7	Write Floor Height data	
	8	Exit SET Mode	7 Segment code display in accordance with MON The settings of the switch are displayed
	C	Check Overload buzzer	Show "OL"
	D	Learning Magnetic Poles	
	F	Tsd Run Check	
1	0-f	Adjustment SHIFT	The setting value is SET0 Current value, press down SW1 Save This value to Ram, the upper and lower lines are exceeded, the value is increased, and the value is reduced
2	0-f	Adjustment Dnsh	The setting value is SET0 Current value, press down SW1 Save This value to Ram If the is exceeded, the value is increased,

			otherwise the value is reduced
3	0-f	Adjustment Wghg	The setting value is SET0 Current value, press down SW1 Save this value to Ram
4	0-f	Adjustment Wgho	The setting value is SET0 Current value, press down SW1 Save this value to Ram
6	0	Opened/Remove the door and flatten the layer Rsrl Function	Originally opened Rsrl function to remove the function, the ' function is ' not opened.
	1	Opened/Cut the door and flatten the floor. Dcrl Function	Originally opened Dcrl function to remove the function, the ' function is ' not opened.
	2	Learn the no-load scale NL	Before entering the "mode", the elevator must be unloaded at the midpoint of the "the" "path.
	3	Learn to balance load scales BL	Before entering this mode, the elevator must balance the "load to stop" at the midpoint of the "the" path
	4	Opened/Cut cable compensation CABLE compensation Function	The original cable compensation function is. To remove the function; The original function is not opened

Attachment 1.1ehy-ii (No Energy Feedback,C Language) Small Classification failure

7seg3	7seg2	7seg1														
		F	E	D	C	B	A	9	8	7	6	5	4	3	2	1
1 Inverter	0 EST	Sd_slit Meld Over load		Sd_ocr Run over Load								Ss_gdf E1 Door Loop Anomaly				Ss_ioc Hardware over Stream
	1 SFS	Sd_hrt Manual over Load		Sw_socr Over load walking line					Sw_fthm2 Heat Sink Temperature Anomaly						Sy_destr Ipc Purge command	Ss_ifo INV Protection Loop Action
	2 NFS		Sw_socr Run over Load	Sw_solr Run over Load					Sw_fthm2 Heat Sink Temperature Anomaly							
	3 NRS		Sd_ocr Run over Load	Sd_olr Run over Load (Regenerative electricity Resistance)					Sw_fthm2 Heat Sink Temperature Anomaly						Sd_cvee Current detection Point drift fault	Sd_cter dc-ct Fault
	4 ALT															
2 Rectifier	0 EST												Ss_cov Cnv Over voltage		Sw_cver Cnv Charge abnormality	
	1 SFS															Ss_clv Cnv Voltage is too low
	2 NFS															
	3 NRS											Sw_dclv Cnv Low voltage		Sw_cvck The charge is over, abnormal check.		Sw_chglt The charge ' s over.
	4 ALT															
3	0 EST															
	1 SFS															
	2 NFS															
	3 NRS															
	4 ALT															
4 Power	0 EST															
	1 SFS													Ss_p12vf 12V Power exception		Ss_pwf E1 Board Power Abnormal

	2 NFS															
	3 NRS															
	4 ALT															St_rambak Ram Backup start
5 Electrical relays	0 EST															
	1 SFS				Sw_cflba #LB on exception	sw_cf5a #5 On exception	sw_cfbk2n #BK2 Forced Off Abnormal	SW_CFBK2B #BK2 off Abnormal	SW_CFBK2A #BK2 on Abnormal	sw_cfbk1n #BK1 Forced Off Abnormal	SW_CFBK1B #BK1 off Abnormal	SW_CFBK1A #BK1 on Abnormal	Sw_cflbn #LB Forced Off Abnormal	SW_CFLBB #LB off Abnormal	sw_cf5n #5 Forced Off Abnormal	sw_cf5b #5 off Abnormal
	2 NFS															
	3 NRS		SW_DFRLB #RL off Fault	Sw_dflra #RL on Fault	sw_dfbk2n #BK2 Forced Off Fault	SW_DFBK2B #BK2 off Fault	SW_DFBK2A #BK2 on Fault	sw_dfbk1n #BK1 Forced Off Fault	SW_DFBK1B #BK1 off Fault	SW_DFBK1A #BK1 on Fault	Sw_dflbn #LB Forced Off Fault	SW_DFLBB #LB off Fault	Sw_dflba #LB on Fault	sw_df5n #5 Forced Off Fault	sw_df5b #5 off Fault	SW_DF5A #5 on Fault
	4 Alt										Sw_cflbn #RL Off fault	Sw_cflra #RL on Fault		Sw_cfbk1a #BK1 on Fault	Sw_cflba #LB on Fault	sw_cf5a #5 on Fault

7seg3	7seg2	7seg1														
		F	E	D	C	B	A	9	8	7	6	5	4	3	2	1
6 Electrical relays	0 EST															
	1 SFS															sw_60e 60 Or 60A Fault
	2 NFS															
	3 NRS								Sw_rssdcia Car Top(后) SSD on Fault	Sw_rssucia Car Top(后) SSU on Fault	Sw_fssdcia Car Top(前) SSD on Fault	Sw_fssucia Car Top(前) SSU on Fault	Sw_ssda Control cabinet SSD on Fault	Sw_ssua Control cabinet SSU on Fault	Sw_hdrnb hdrn off Fault	Sw_hdrna HDRN on Fault
	4 ALT															
7 Safety Loop	0 EST								Lamd_test Maintenance Automatic Low- speed operation							
	1 SFS	Sn_29 #29 Safety Circuit						Sfs_test Maintenance Sfs Test	Sp_bkzn Braking torque adjustment interval stop					Sw_est Est Action 2 次	Sc_s29 Managing Software Exceptions	Sd_32gq Driver Software exception

	2 NFS															
	3 NRS						Mnt_nrs Maintenance operation NRS			Sw_ndpd Departure anomaly	Sc_nrslrq Management can no longer be started	Sw_est2 E-STOP2 次	Sw_est1 E-STOP1 次	Sd_dnrs Dr Side cannot reboot	Sw_d89 #89 回路断开	
	4 ALT	Sc_gcalt Group Management s/w	Sc_msalt Management s/w	Sy_sfrst Security Loop Reset Command									Sw_est1 E-stop1 次			
8 Door	0 EST															
	1 SFS													Ss_dkc Open the door to run (SLC)	Sw_dopn Open the door in operation	
	2 Nfs								sw_rsdel2b Backdoor SDE off Fault	sw_fsdel2b Front DoorSDE off Fault	sw_rclt12b Backdoor CLT off Fault	sw_fcilt12b Front Door CLT off Fault	SW_41DGB 41DG off Fault			
	3 NRS												sw_rolt12b Backdoor OLT off Fault	sw_folt12b Front Door OLT off Fault	Sw_41dga 41DG on Abnormal	
	4 ALT			sw_droilt12b Backdoor OLT off Fault	sw_droilt12a Backdoor OLT on Fault	sw_dfolt12b Front Door OLT off Fault	sw_dfolt12a Front Door OLT on Fault	sw_drfg12b Backdoor FG off Fault	sw_drfg12a Backdoor FG on Fault	sw_dffg12b Front Door FG off Fault	sw_dffg12a Front Door FG on Fault	sw_drcilt12b Backdoor CLT off Fault	sw_drcilt12a Backdoor CLT on Fault	sw_dfclt12b Front Door CLT off Fault	sw_dfclt12a Front Door CLT on Fault	Sw_dsopn Layer Station Ds Open Door Check out (EN81)
9 Shaft	0 EST															
	1 SFS	Sw_ulc Uplink U1 Switch Action	Sw_dlc Downlink DL Switch Action	SW_UDLB UL/DL All action					Sw_rrldb Backdoor RLD off Fault	Sw_rrlub Backdoor RLU off Fault	Sw_frldb Front Door RLD off Fault	Sw_frlub Front Door RLU off Fault	Sw_ts3se Tsd Switch failure			
	2 NFS															
	3 NRS	SW_UDLB UL/DL All action	SW_UDSRB USR/DSR All action	Sw_rdzrlx Abnormal switch of back door flat layer	Sw_fdzrlx Abnormal switch of front door flat layer	Sw_dzccab Dz Detection circuit On/off Fault		Sw_rrlda Backdoor RLD on Fault	Sw_rrlua Backdoor RLU on Fault	Sw_rdzda Backdoor DZD on Fault	Sw_rdzua Backdoor Dzu on Fault	Sw_frlda Front Door RLD on Fault	Sw_frlua Front Door RLU on Fault	Sw_fdzda Front Door DZD on Fault	Sw_fdzua Front Door Dzu on Fault	
	4 ALT				Sw_fmcte Floor Count Anomaly											
A Traction System	0 EST															
	1 SFS					SW_CFBK2N #BK2 Forced Off		sw_cfbkln #BK1 Forced Off								
	2 NFS													Sw_bker Brake exception	Sw_mthm Abnormal temperature of motor	

3 NRS		Sd_azer PM Z Phase Anomaly	Sd_thng Magnetic Pole Not Learning											Sw_bker Brake exception	Sw_mthm Abnormal temperature of motor
4 ALT															

7seg3	7seg2	7seg1														
		F	E	D	C	B	A	9	8	7	6	5	4	3	2	1
B Speed	0 EST	Ss_ast Slc Stall	Ss_tgbh Slc Too High Speed	Ss_dest Slc Urgent stop					Sw_astw Traction Machine Plug-around	Sw_tgr1 Flat Layer Anomaly		Sw_tgbx Encoder deviation	Sw_ast Wire Rope Skid	Sw_tgbr Retrograde	Sw_tgbh Too High Speed	Sw_tgbl Too Low
	1 SFS								Sw_los Manual over High speed				Sw_lovr Too much travel	Sw_dect Deceleration time limit	Sd_pvjp Speed and graphics deviation fault	Sw_tgbl Too Low
	2 NFS													Sw_dectn Deceleration time limit(Slow stop)		Sw_rlt Re-leveling time limit
	3 NRS								Sw_fmser Fms Failed	Sw_tgr1 Flat Layer Anomaly		Sw_ast Wire Rope Skid	Sw_tsak Tsd Run	Sw_seld Layer Picker Failure	St_lsa Run detection	Sd_ovjp Comparison of speed and graphics (times)
	4 ALT										St_tlprn TLP Walking Check out			St_selz Separator Deviation Fault	St_slzq Selector bias Warning	
C 秤	0 EST															
	1 SFS															
	2 NFS													Sw_wgcn The weighing device is disconnected		Sw_wge2 Run the scale anomaly
	3 NRS												Sw_wgwe Scale Set Abnormal	Sw_wgcn The weighing device is disconnected	Sw_wge1 Stop the scale anomaly	Sw_wge2 Run the scale anomaly
	4 ALT															
D DR	0 EST													Ss_ast Slc Stall	Ss_tgbh Slc Too High Speed	Ss_dest Slc Urgent stop
	1 SFS				SS_CCWC4 CC_WDT4 次	SS_CCWC5 CC_WDT5 次							Ss_dr_d89 Dr D89 Instructions	Ss_dr_d5 DR D5 Instructions	Ss_tsdng SLC TSD Abnormal	Ss_dkc Slc Open the door to run

	2 NFS															
	3 NRS			SS_CCWC3 CC_WDT3 Times												Sd_mcpwde DR WDT Fault
	4 ALT	Ss_ccwc1 Cc_wdt1 Times	Ss_ccwc2 Cc_wdt2 Times				Ss_cct10s Cc_wdt Shielding Timer Action									Ss_mcpt10s DR WDT Shielding Timer Action
E System Setup	0 EST															
	1 SFS															
	2 NFS															
	3 NRS						Sw_ramer Control Ram Fault		Sw_ssyer Speed setting failure	Sw_tsyer Setting Fault of Traction machine	Sw_asyser Acc Code setting failure		Sd_dter Drive System Setup Failure	Sd_ramer Driven Ram Fault		
	4 ALT															
F Communication	0 EST															
	1 SFS															
	2 NFS															
	3 NRS			Sw_ster_rdc2 Backdoor IC Failure	Sw_ster_rcs2 Front Door IC Fault	Sw_ster_fdc2 Back door BC1 fault	Sw_ster_fcs2 Front Door BC1 Fault	Sw_ster_wgd Scale failure	Sw_ster_rdc1 Backdoor DC Fault	Sw_ster_rcs1 Backdoor Cs Fault	Sw_ster_fdc1 Front Door DC Fault	Sw_ster_fcs1 Front Door Cs Fault	Sw_ster_asc1 Sc Fault			Sw_ster_plc1 Plc1 Fault
	4 ALT								Sw_rdc1_hng Backdoor DC Fault	Sw_rcs1_hng Backdoor Cs Fault	Sw_fdc1_hng Front Door DC Fault	Sw_fcs1_hng Front Door Cs Fault	Sw_asc1_hng Sc Fault			Sw_plc1_hng Plc1 Fault
	5 ALT2					Sw_rbc4_hng Back door BC4 fault	Sw_rbc3_hng Back door BC3 fault	Sw_rbc2_hng Back door BC2 fault	Sw_rbc1_hng Back door BC1 fault	Sw_fbc4_hng Front Door BC4 Fault	Sw_fbc3_hng Front Door BC3 Fault	Sw_fbc2_hng Front Door BC2 Fault	Sw_fbc1_hng Front Door BC1 Fault	Sw_fic5_hng Backdoor IC Failure	Sw_fic1_hng Front Door IC Fault	Sw_hsal1_down_hng HS Fault